

Department of Computer Science & Engineering

CS601: Applied Artificial Intelligence & RAG Systems

Semester: Fall 2026 | **Instructor:** Dr. Rajesh Sharma | **Credits:** 4.0

Class Schedule: Mondays and Wednesdays, 10:00 AM – 11:30 AM (Room LH-302)

Prerequisites: Proficiency in Python programming, linear algebra, and basic probability theory.

1. Course Description & Learning Objectives

This course provides an end-to-end practical understanding of Modern Artificial Intelligence, with a core focus on Large Language Models (LLMs), Embeddings, Vector Databases, and Retrieval-Augmented Generation (RAG). By the end of this course, students will be able to design, implement, and deploy production-ready RAG systems that eliminate hallucinations and ground LLM responses in proprietary knowledge bases.

2. Course Modules & Detailed Syllabus

Module 1: Foundations of NLP and Tokenization

Word embeddings (Word2Vec, GloVe), subword tokenization (BPE, WordPiece), Transformer self-attention mechanism, and encoder-decoder architectures. Practical exercises include tokenization with Hugging Face.

Module 2: Large Language Models & Prompt Engineering

Autoregressive generation, zero-shot vs few-shot learning, chain-of-thought prompting, temperature and top-p sampling. Students will work with models like Google Gemini, GPT-4, and LLaMA.

Module 3: Retrieval-Augmented Generation (RAG) Architecture

The fundamental limitations of parametric LLM memory: hallucination, cutoff dates, and domain ignorance. The RAG paradigm: Document loaders (PyPDFLoader, Unstructured), chunking strategies (recursive character splitting, token-based, semantic chunking), vector embeddings (Google text-embedding-004, sentence-transformers MiniLM), vector stores (FAISS, Chroma, Pinecone), and top-k similarity retrieval algorithms (cosine similarity, inner product, L2 distance).

Module 4: Evaluation, Grounding, and Deployment

RAG evaluation metrics: Context Precision, Context Recall, Faithfulness, and Answer Relevance. Deployment using Gradio, Streamlit, and FastAPI. Edge cases: empty documents, multi-file retrieval, and latency optimization.

3. Grading Policy & Assessment Scheme

The overall course assessment consists of five components:

Assessment Component	Weightage	Due Date
Lab Assignments (4 in total)	20%	Fortnightly
Midterm Examination	25%	Week 8 (October 24)
Capstone RAG Project & Demo	35%	Week 14 (December 2)
Final Viva Voce	10%	Week 15 (December 10)
Class Participation & Quizzes	10%	Ongoing

4. Capstone Project Requirements

Students must build a complete RAG system with a Gradio web interface named 'Ask My Documents – RAG Assistant'. The project must demonstrate document loading with PyPDFLoader, chunking with RecursiveCharacterTextSplitter, embedding generation, FAISS indexing, similarity retrieval, and strict grounded answer generation with source citation. Late project submissions will incur a 10% penalty per calendar day up to a maximum of 3 days, after which submissions will not be accepted.